

SILBO RUC65 5G Cellular Router

Secure 5G Cellular with High-Speed WiFi 6, AX3000 and Dual-SIM PoE/DC Powered

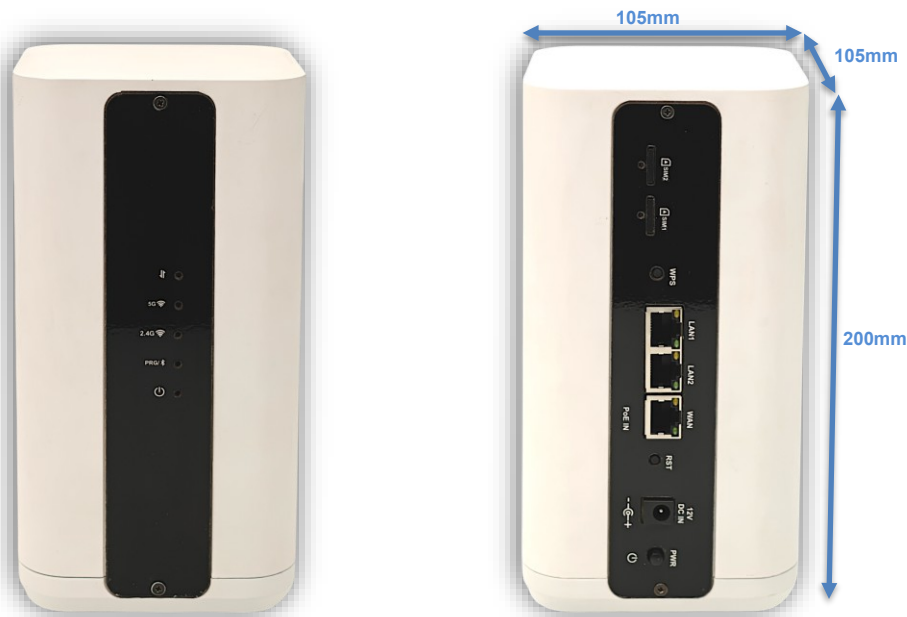
The SILBO RUC65 is a compact, enterprise-grade 5G cellular router designed to provide blazing-fast connectivity for modern commercial, industrial, and remote office environments. Powered by the robust MediaTek MT7981 networking processor and integrated with advanced WiFi 6 and ultra-low-latency Bluetooth 6.0 (BLE) capabilities, the RUC65 offers high-speed data delivery, dual-SIM redundancy, and versatile power options including PoE IN. Housed in a fanless desktop enclosure, it serves as an ideal edge gateway for mission-critical operations requiring maximum network uptime.

Key Highlights

- **Ultra-Fast 5G Sub-6 Connectivity:** Delivers up to 2.2 Gbps download speeds with fallback to high-speed 4G LTE, ensuring continuous operation even in dynamic signal environments.
- **Dual-Band WiFi 6 Performance:** Full MIMO 802.11ax support across 2.4GHz and 5GHz bands with up to 160MHz channel bandwidth for dense device environments.
- **Next-Gen Bluetooth 6.0 Integration:** Features the advanced Nordic nRF54L05 BLE 6.0 SoC acting as a dedicated LE HCI Adapter for Linux BlueZ host stack offloading, coupled with a powerful Range Extender (FEM) for long-range IoT sensor communication.
- **Dual SIM Failover Redundancy(Optional eSIM):** Smart, automated SIM switching based on signal strength, data limits, or ping monitoring guarantees persistent network uptime.
- **Flexible Power Architecture:** Supports standard 12V DC input alongside IEEE 802.3at PoE IN over WAN, making it adaptable for optimal positioning without nearby power outlets.
- **OpenWrt Extensibility:** Built on an OpenWrt Linux baseline, allowing for custom shell scripting, C/C++ packages, and seamless integration with custom enterprise firmware.

Product Image





Technical Specifications

Parameter	Specification
CPU	MediaTek MT7981
System Memory	512 MB DDR3 RAM
Storage (Flash)	128 MB NAND
Ethernet Interfaces	1x 10/100/1000 WAN (PoE IN IEEE802.3at), 2x 10/100/1000 LAN
SIM Slots	2x Nano SIM (Active / Passive configuration)
Antenna Design	Fully Internal (Cellular, Wi-Fi, and BLE)
Physical Controls	Pinhole Reset Button, WPS Button
Status LEDs	1x Power, 1x Wi-Fi 5G, 1x Wi-Fi 4G, 1x PRG/BLE, 1x 5G Mode, 2x SIM, 3x Ethernet
Hardware Fail-safe	Hardware Watchdog, Dying Gasp notification support(Optional)
Enclosure / Mounting	Plastic Housing, Fanless Passive Cooling / Table Top Mounting

Cellular, Wi-Fi & BLE Specifications

Parameter	Specification
Cellular Module & Speeds	5G SA Sub-6: Up to 2 Gbps (DL) / 1 Gbps (UL) 5G NSA Sub-6: Up to 2.2 Gbps (DL) / 575 Mbps (UL) 4G LTE: Up to 600 Mbps (DL) / 150 Mbps (UL) 3G: Up to 42.2 Mbps (DL) / 11 Mbps (UL)
Modem Diagnostics	Signal Strength (RSSI), SINR, RSRP, RSRQ tracking
Band / APN Management	Auto / User configurable Band Management, Automatic / Manual Operator Lock, Auto APN support
SIM Configuration & Auth	Single SIM, Dual SIM modes; Signal strength, ICMP echo, or Data limit based SIM switching; PAP/CHAP/No Authentication
SMS Capabilities	Send SMS, Read SMS, Administrative SMS Commands
Wi-Fi Technology	Wi-Fi 6 Dual-Band MIMO 802.11 a/b/g/n/ac/ax RF Bandwidth: 20 / 40 / 80 / 160 MHz
Wi-Fi Modes & Security	Access Point (AP), Station, AP & Client, Guest Wi-Fi, Mesh Wi-Fi Authentication: WPA/WPA2/WPA3 (PSK) & Mixed Modes Encryption: TKIP, AES, TKIP & AES
Wi-Fi Management	Configurable TX Power, Scheduled ON/OFF Scheduler, WMM, MAC ID Filtering (Whitelisting/Blacklisting)
Bluetooth (LE) Integration(Optional)	Bluetooth 6.0 via Nordic nRF54L05 BLE SoC Features 128 MHz ARM Cortex-M33 processor, 500KB NVM, 96KB RAM
Bluetooth Mode	Dedicated LE HCI Adapter for Linux BlueZ host stack offloading
BLE Front-End Module	nRF21540 Range Extender with integrated +20dBm PA and 13dB LNA Gain Operating Frequency: 2400 MHz to 2500 MHz, 1MHz channel spacing
BLE Performance	On-the-air Data Rate: 125 kbps - 4000 kbps Transmit Power: 7dBm at BLE SoC (Additional 5dBm to 22dBm via FEM PA) Receive Sensitivity: -104 dBm @125 kbps (Bluetooth LE mode)

Software Features

Parameter	Specification
Routing & Protocols	Static routing, Advanced Static Routing; TCP, UDP, IPv4, IPv6, ICMP, NTP, HTTP, HTTPS, SCP, PPP, SSH, DHCP (Static & Dynamic)
Connection Monitoring	Modem & Router monitoring with ICMP Echo, Ping Reboot, Scheduled Periodic Reboot, Hardware Watchdog (Enable/Disable)
Firewall Features	Port forward, traffic rules, SNAT, Zone forwarding, Parental Control, Masquerading, MSS Clamping, Loopback IP, Loopback Rules
Attack Prevention	SYN Flood Protection, Drop Invalid Packets
Multi-WAN Configuration	Configurable Failover and Load Balancing between Cellular WAN, Wi-Fi WAN, and Wired WAN
VLAN Support	Tagged VLAN and Untagged VLAN routing segmentation
Robust Tunneling & VPN	OpenVPN: TUN/TAP with TLS Certificates IPsec: IKEv1, IKEv2, Aggressive/Main Mode; Encryption: 3DES, AES128, AES192, AES256; Authentication: SHA1, MD5, SHA256, SHA384, SHA512; DH Group: MODP768 to MODP4096 Additional Protocols: WireGuard, ZeroTier, GRE, L2TP, PPTP (with/without MPPE), IPIP Tunnel
System & Diagnostics	Open-WRT based Linux OS supporting Shell scripting, C, C++; SDK environment package provided (Optional). Built-in Ping, Traceroute, Modem Debug Info, Iperf, Download Logs, MAC binding, Router Health Monitoring, Package Manager

Parameter	Specification
NMS	Remote Monitoring & Management Platform unified deployment for SILBO Networking products
NMS VPN	Secure Remote Access of localized SILBO and non-SILBO Networking edge devices
CPE Management	TR069 Client integration for seamless automated Service Provider provisioning

Operational, Environmental & Packaging

Parameter	Specification
Input Voltage / Connection	12 Vdc Input via DC Jack OR PoE IN (IEEE802.3at) via WAN Port
Operating Temperature	0 to 50 deg C
Storage Temperature	-20 to 60 deg C
Environmental Humidity	10% ~ 90% (non-condensing)
Dimensions / Weight	105 x 200 x 105 mm (W x H x D) / 590 grams
Standard Packaging Content	RUC65 Cellular Router, Power Adapter, 1-meter RJ45 Ethernet cable

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